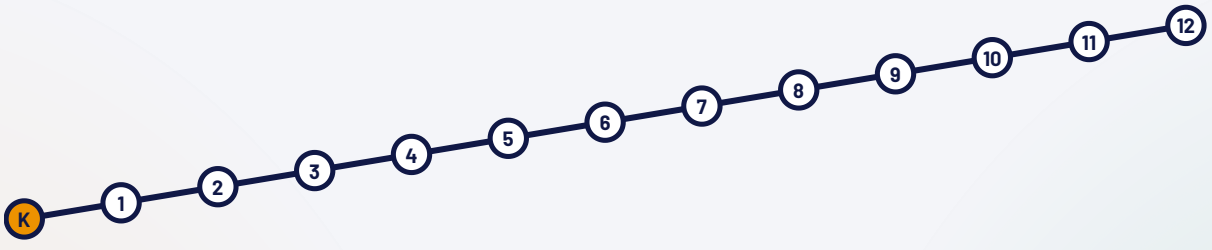
Awsaj Academy

K-12 VERTICAL ALIGNMENT · AN ANALYTICAL WALKTHROUGH

Our Curriculum Journey: Science

What our teachers' ratings tell us about how our standards are taught, from Kindergarten to Grade 12, with questions and options for our part in the story.



These findings come from the ratings you entered as teachers. They show what teachers reported as taught across each grade (2,709 ratings, completed November 2026), not what students know or can do, and they do not judge any teacher. Every item links to the data on the Vertical Alignment site.

Contents

1	**Science, K-12** Science from Kindergarten to Grade 12.	3
2	**About the data (at the back)** Where the data comes from, how to read the ratings, the words we use, and the limits.	15

Each part follows the same path: the picture (charts), what the numbers say, what we notice, questions for us, and our part in the story (options for improving the curriculum, working across grade levels, knowing the bands above and below, and a concrete action). The data behind each item is shown in brackets, and each item links to the matching page of the Vertical Alignment site. About the data (where the data comes from, how to read the ratings, the words we use and the limits) is at the back. 18 pages, A4. The bookmarks jump to each part.

Taught in depth Introduced / exposed Not taught

SCIENCE, KINDERGARTEN TO GRADE 12

Science, K-12

Across K-12, teachers reported that 35% of the 247 Science ratings (208 standards) are taught in depth, level with ELA as the highest in the school, and 58% are at least introduced. High school Physical Science (88% in depth) is one of the strongest strands in any subject.

35%
taught in depth

58%
at least introduced

42%
not taught

247
grade-level ratings

What you are seeing. Four headline figures for this part: the share taught in depth, at least introduced and not taught, and how many ratings they come from.

What it adds to our story. This is the starting picture for this part. The slides that follow show where these numbers come from and what they might mean for us.

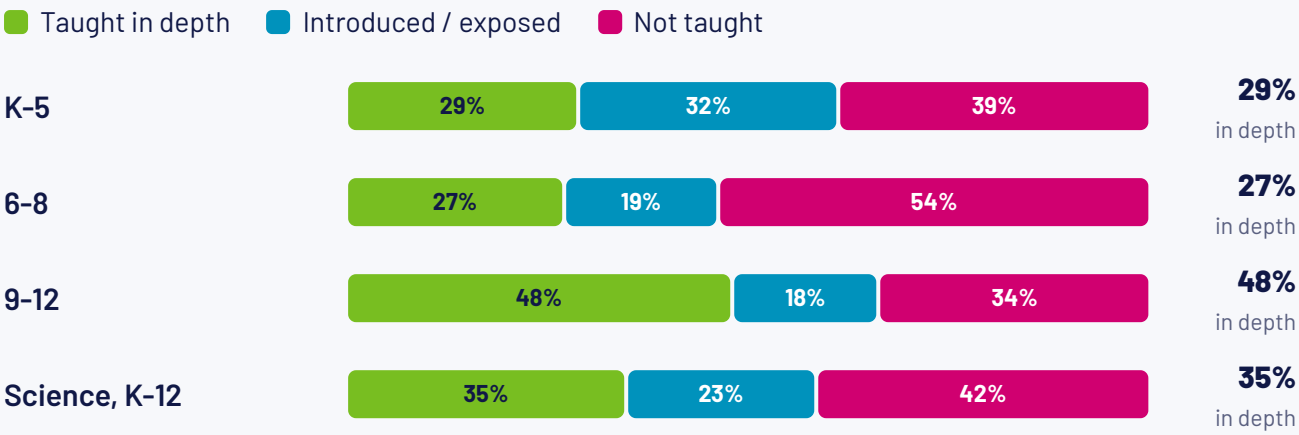
These findings come from the ratings you entered as teachers. They show what teachers reported as taught across each grade, not what students know or can do. We are sharing them back so we can see the whole picture together and decide how we move forward. Each part ends with options for our part in the story. Every item links to the data on the Vertical Alignment site.

THE PICTURE

At a glance

What you are seeing. Each bar shows all the ratings for one subject: green is taught in depth, blue is introduced, pink is not taught.

What it adds to our story. Science is often all or nothing: a topic is either taught deeply or not taught that year.



How to read this: each bar is 100% of the grade-level ratings. Green is taught in depth, blue is introduced or exposed, pink is not taught. Essential Elements are counted apart.

Open the coverage map on the site ↗

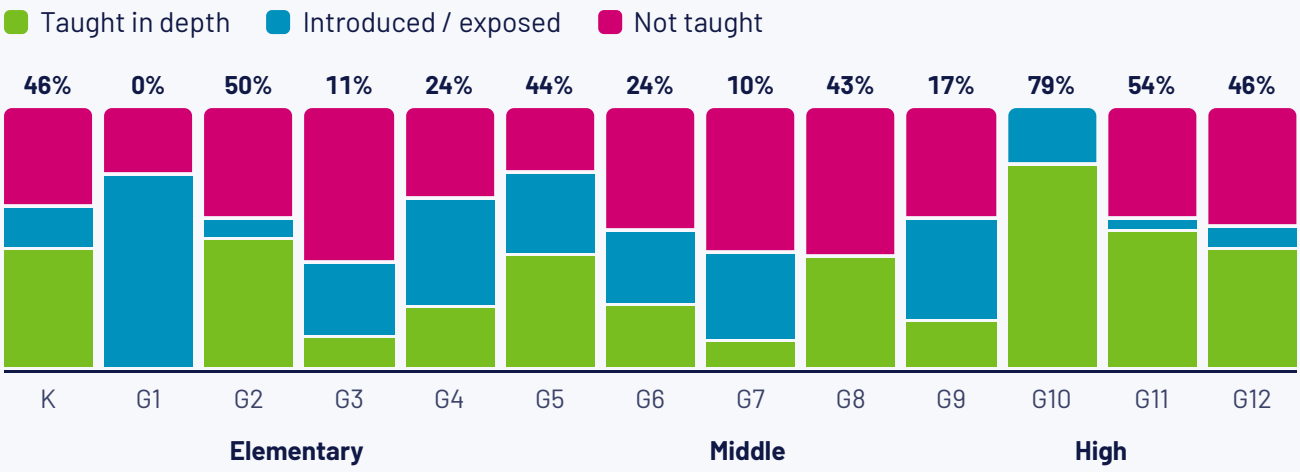
	Rated	In depth	Introduced	Not taught
K-5 (AERO Science)	**90**	**29%**	**32%**	**39%**
6-8 (NGSS)	**74**	**27%**	**19%**	**54%**
9-12 (NGSS)	**83**	**48%**	**18%**	**34%**
K-12	**247**	**35%**	**23%**	**42%**

THE PICTURE

Grade by grade

What you are seeing. One column for each grade. The number on top is the share of that grade's standards taught in depth.

What it adds to our story. Depth changes from year to year, which may simply follow which units each grade runs.

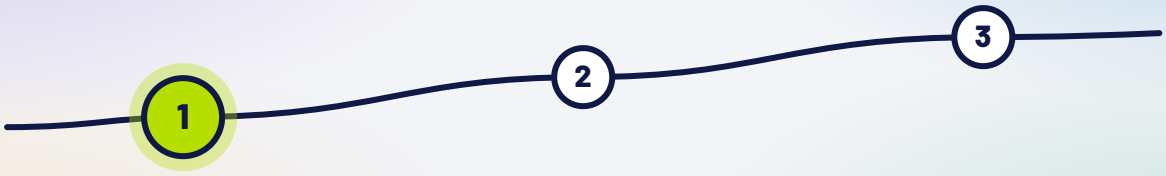


How to read this: one column per grade. The number on top is the share taught in depth.

Strand	K	G1	G2	G3	G4	G5	G6	G7	G8	G9	G10	G11	G12
Earth and Space Science	40%	0%	100%	0%	0%	0%	67%	0%	0%	21%			
Engineering, Technology, and Applications of Science	33%	0%	0%	0%	0%	33%	0%	0%	100%	0%	0%	75%	0%
Life Science	100%	0%	100%	25%	100%	50%	0%	15%	0%			50%	
Physical Science	50%	0%	0%	0%	29%	83%	0%	0%	62%		100%		67%

How to read this: each square is the share of that strand's rated standards taught in depth in that grade. Darker green means more in depth. A blank square means the grade has no standards in that strand.

Open the grade dashboards ↗ Gaps and repetition ↗ Progressions ↗



SECTION 1 OF 3

Key facts

What you are seeing. The main facts from the ratings, in plain numbers. Each one links to the data on the Vertical Alignment site.

What it adds to our story. When Science is taught, it is taught deeply. The open question is which year each topic lives in.

1. THE BIG PICTURE **Depth by grade.** K 46%, G1 0%, G2 50%, G3 11%, G4 24%, G5 44%, G6 24%, G7 10%, G8 43%, G9 17%, G10 79%, G11 54%, G12 46%. Coverage map ↗
2. THE BIG PICTURE **Few Science ratings are introduced.** 23% of Science ratings are introduced, about half the rate in Math (45%) and ELA (44%). Grade 8 has none at all. Grade 8 dashboard ↗
3. STRANDS AND TOPICS **Grade 10 Science.** 79% in depth, 21% introduced, nothing not taught. All 15 Physical Science standards are in depth. Grade 10 dashboard ↗
4. STRANDS AND TOPICS **Strands K-12, share in depth.** Physical Science 52%, Life Science 33%, Earth and Space 28%, Engineering 20%. Coverage map ↗
5. STRANDS AND TOPICS **Physical Science by band.** 31% in K-5, 40% in 6-8, 88% in 9-12. Coverage map ↗
6. STRANDS AND TOPICS **Life Science by band.** 47% in K-5, 7% in 6-8, 50% in 9-12. Coverage map ↗
7. STRANDS AND TOPICS **Earth and Space by band.** 25% in K-5, 40% in 6-8 (67% in Grade 6), 21% in 9-12. Grade 6 dashboard ↗
8. STRANDS AND TOPICS **Engineering.** 20% in depth K-12, with 2 bright spots: Grade 8 (100%) and Grade 11 (75%). Coverage map ↗

9. ACROSS GRADES AND SCHOOLS **Progressions.** 7 of 40 progressions break down on an official link, all in K-5. In K-5, 11 of 31 progressions are taught at every step, the most of any K-5 subject. Progressions ↗

10. ACROSS GRADES AND SCHOOLS **Hand-overs.** 17 standards taught in depth in Grades 6 to 10 build on a K-5 standard rated not taught: 11 land in middle school and 6 in high school. All rest on suggested links, not official ones, so each is worth checking first. Progressions ↗

11. PRIORITY AND POWER STANDARDS **Priority and power standards.** 48% in depth, the highest of any subject: 37% in K-5, 50% in 6-8, 55% in 9-12. Grade 8 priorities are 85% in depth. Priority and power (Audit) ↗

12. ESSENTIAL ELEMENTS **Essential Elements.** 36 ratings, in Grades 6 to 12 only: 44% in depth. K-5 uses AERO Science, which has no Essential Elements. Coverage map ↗

13. GAPS AND PLACEMENT **Standards no grade teaches.** 24 confirmed, all in K-5: Physical Science 12, Earth and Space 8, Engineering 2, Life Science 2. Gaps and repetition ↗

14. GAPS AND PLACEMENT **Not taught where placed.** 49 standards in Grades 6 to 12 (29 in middle school, 20 in high school), where grade placement was a suggestion. Gaps and repetition ↗

15. GAPS AND PLACEMENT **Introduced, never in depth.** 49 standards are introduced and never in depth, the fewest of any subject: 28 in K-5, 10 in 6-8, 11 in 9-12. Gaps and repetition ↗



SECTION 2 OF 3

Relevant patterns

What you are seeing. What we notice when we put the facts side by side: rises, dips and comparisons. These are possibilities to talk about, not conclusions.

What it adds to our story. Much of "not taught" in secondary Science may be a placement question, and Science also carries a lot of literacy.

1. THE BIG PICTURE **Science is often all or nothing.** A standard is usually either in depth or not taught, with little in between (35% in depth, 42% not taught, 23% introduced). One possible reason is that Science is planned in whole units: a topic is either a unit that year or it is not. Grades 1, 4 and 9 are exceptions, with many standards introduced. Coverage map ↗

2. THE BIG PICTURE **Grade 9 is the quiet year in high school.** 17% in depth, between Grade 8 (43%) and Grade 10 (79%). Coverage map ↗

3. STRANDS AND TOPICS **Life Science has a long quiet stretch.** Strong in K-5 (100% in Grades 2 and 4), then 0% in Grades 6 and 8 and 15% in Grade 7, before returning in Grade 11 (50%). Students may go several years between deep Life Science units. Coverage map ↗

4. STRANDS AND TOPICS **Earth and Space is spread thin.** It is the strand with the most "introduced only" standards (22), including nearly half of Grade 9's 19 standards. Gaps and repetition ↗

5. STRANDS AND TOPICS **Engineering is taught in bursts.** It is shared across grade bands (K-2, 3-5, middle, high) and reaches depth in only a few grades. It may be a strand no single grade feels it owns. Coverage map ↗

6. STRANDS AND TOPICS **Grade 1 is the exception to the pattern.** 75% introduced and none yet rated in depth, with only Engineering not taught. Grade 1 appears to touch Science widely and lightly. Coverage map ↗

7. ACROSS GRADES AND SCHOOLS **That is a strength with a consequence.** When Science is taught here, it is taught deeply (Science has the fewest "introduced only" standards: 49, against 105 to 241 in the other subjects). The other side is that coverage depends entirely on which units each grade runs. Gaps and repetition ↗

8. ACROSS GRADES AND SCHOOLS **Depth alternates from year to year.** High, low, high, low: 46%, 0%, 50%, 11% in Kindergarten to Grade 3, and 24%, 10%, 43%, 17%, 79% from Grade 6 to Grade 10. Students meet Science in strong years and lighter years. A unit-based plan would produce this pattern, which teams could confirm. Coverage map ↗

9. ACROSS GRADES AND SCHOOLS **Physical Science is a through-line worth protecting.** It builds from 31% to 40% to 88%. But 12 K-5 Physical Science standards are taught by no grade, and 9 hand-over gaps run from those into Grade 8 and Grade 10 Physical Science. Progressions ↗

10. ACROSS GRADES AND SCHOOLS **K-5 progressions are the healthiest in the band.** Eleven K-5 Science progressions are taught at every step. Where the units exist, they connect. Progressions ↗

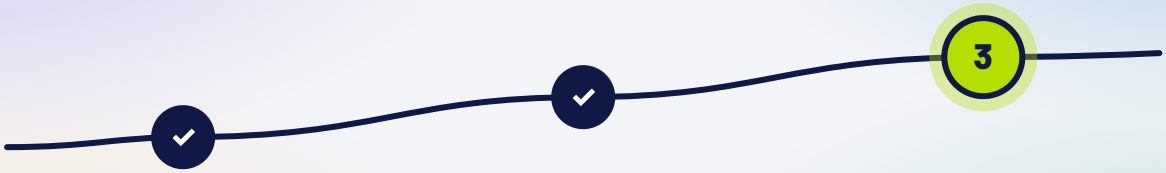
11. PRIORITY AND POWER STANDARDS **Priority standards are deeper than Science overall.** Science priority standards are in depth more often than Science overall in every band (37%, 50%, 55%). Grade 7 is the exception (2 of 14). Priority and power (Audit) ↗

12. GAPS AND PLACEMENT **Each secondary grade seems to own a discipline.** Grade 6 Earth and Space, Grade 8 Physical Science and Engineering, Grade 9 Earth and Space, Grade 10 Physical Science, Grade 11 Life Science, and Grade 12 Physical Science (6 of 9 in depth). Grade 7 holds most of the middle school Life Science standards, mostly introduced. Standards placed in a grade that teaches a different discipline show as not taught. Coverage map ↗

13. GAPS AND PLACEMENT **So much of secondary "not taught" is a placement question.** There are no confirmed Science gaps in Grades 6 to 12. The 49 standards not taught where placed may simply sit in the wrong year on our map. Gaps and repetition ↗

14. CAPACITY AND CHOICES **Capacity shows what a grade carries in Science each year.** A grade takes about 7 Science standards to depth a year, introduces about 4 and does not teach about 8, out of about 19 rated. In K-5 it is about 4 of 15, in middle school 7 of 25, in high school 10 of 21. As shares of Science standards: K-5 29% in depth, 32% introduced, 39% not taught; 6-8 27% in depth, 19% introduced, 54% not taught; 9-12 48% in depth, 18% introduced, 34% not taught. Capacity here means how many standards a grade's teaching time actually holds in a year. It is not a target and not a judgement. Coverage map ↗

15. LITERACY ACROSS SUBJECTS **Science is also a literacy subject.** Science units teach vocabulary, informational reading and explanation writing. Teams may be teaching ELA standards in Science that only the ELA ratings would show. Coverage map ↗



SECTION 3 OF 3

Where do we go next?

What you are seeing. Questions for teams to talk about. Each one shows the data that prompted it, and links to where that figure appears on this page.

What it adds to our story. These questions are about a shared K-12 map of Science units, so every strand has a home.

1 **STRANDS AND TOPICS** What makes Grade 10 Physical Science work so well, and what could other courses borrow from it?

(Data: Grade 10 Science is 79% in depth with nothing not taught; all 15 Physical Science standards are in depth) see Key fact 3

Coverage map ↗

2 **STRANDS AND TOPICS** Who owns Engineering in each band, and could it be woven into existing units?

(Data: Engineering is 20% in depth K-12, with depth only in Kindergarten, Grade 5, Grade 8 and Grade 11) see Key fact 8

Coverage map ↗

3 **STRANDS AND TOPICS** What would turn Grade 9 Earth and Space from introduced to in depth?

(Data: Grade 9 Earth and Space is 21% in depth across 19 standards; Earth and Space has 22 introduced-only standards K-12, the most of any Science strand) see Pattern 4

Coverage map ↗

4 **STRANDS AND TOPICS** How could Science units carry reading, writing and vocabulary, so one unit serves two subjects?

(Data: Science is 35% in depth K-12; K-5 ELA Language is 10%) see Key fact 1

Coverage map ↗

1 **STRANDS AND TOPICS** From Grade 10 Physical Science, which 2 practices (a shared assessment, reteaching time, a lab routine) could another Science year try next term?

(Data: Science priority standards are 48% in depth, the highest of any subject; 11 K-5 progressions are taught at every step) see Key fact 11

Priority and power (Audit) ↗

2 **STRANDS AND TOPICS** Which ELA vocabulary, reading or writing standards do our Science units already teach, and could we name them in the unit plan?

(Data: Science is 35% in depth K-12; K-5 ELA Language is 10%) see Key fact 1

Coverage map ↗

3 **ACROSS GRADES AND SCHOOLS** Is the year-to-year alternation by design, and what do students experience in a lighter Science year?

(Data: In depth by grade: 46%, 0%, 50%, 11%, 24%, 44% in K-5; 24%, 10%, 43% in 6-8; 17%, 79% in Grades 9 and 10) see Key fact 1

Coverage map ↗

4 **ACROSS GRADES AND SCHOOLS** Which K-5 Physical Science ideas do Grade 8 and Grade 10 most rely on, and where could they be placed?

(Data: 12 K-5 Physical Science standards are taught by no grade; 9 hand-over gaps run from them into Grades 8 and 10) see Key fact 1

Progressions ↗

1 **ACROSS GRADES AND SCHOOLS** How might students keep meeting Life Science between Grade 5 and Grade 11?

(Data: Life Science is 47% in depth in K-5, 0% in Grades 6 and 8, 15% in Grade 7, and 50% in Grade 11) see Pattern 3

Coverage map ↗

2 **ACROSS GRADES AND SCHOOLS** What would Grade 6 like K-5 students to arrive knowing and able to do in Science?

(Data: 11 hand-over gaps run from K-5 into Grades 6 and 8; Grade 5 to Grade 6, Science in depth falls from 44% to 24%) see Key fact 1

Progressions ↗

3 **GAPS AND PLACEMENT** **Do we have a K-12 map of which Science units run in which year, and does it match what the ratings show?**

(Data: Only 23% of Science ratings are introduced; standards are mostly either in depth or not taught) see Key fact 2

Coverage map ↗

4 **GAPS AND PLACEMENT** **Should the standards on the site be moved to the grades that teach each discipline, so the picture is accurate?**

(Data: No confirmed Science gaps in Grades 6 to 12, but 49 standards are not taught where placed) see Key fact 4

Gaps and repetition ↗

1 **GAPS AND PLACEMENT** **For the 24 K-5 standards no grade teaches, which should we place and which do we choose not to use?**

(Data: Physical Science 12, Earth and Space 8, Engineering 2, Life Science 2) see Key fact 13

Gaps and repetition ↗

2 **CAPACITY AND CHOICES** **With about 4 to 10 Science standards in depth per grade, does our unit map spend that depth where it matters most?**

(Data: Per grade: K-5 4 of 15 in depth, 6-8 7 of 25, 9-12 10 of 21) see Pattern 14

Coverage map ↗

3 **CAPACITY AND CHOICES** **Science is mostly in depth or not taught: is that the balance we want, or would we introduce some topics briefly in other years?**

(Data: Science mix: 35% in depth, 23% introduced, 42% not taught) see Pattern 1

Coverage map ↗

Our part in the story

What you are seeing. Options for what we could do, under four headings, each with the data behind it.

What it adds to our story. This is where findings turn into action: small, shared steps we could take together.

Options to discuss and act on. Pick what fits your team, change it, or add your own.

Improving the curriculum

We could draw one K-12 map of which Science units run in which year, so every strand has a planned home.

(Data: Only 23% of Science ratings are introduced; standards are mostly either in depth or not taught) **see Key fact 2**

Coverage map ↗

Working across grade levels

We might ask Grade 10 Physical Science to share what makes its units work with the Grade 4, 5 and 8 teams.

(Data: Grade 10 is 79% in depth; all 15 Physical Science standards are in depth) **see Key fact 3**

Coverage map ↗

Knowing the bands above and below

In K-5, we could place the Physical Science standards that Grade 8 and Grade 10 build on.

(Data: 12 K-5 Physical Science standards taught by no grade; 9 hand-over gaps run into Grades 8 and 10) **see Key fact 1**

Progressions ↗

A concrete action we could take

We might plan a Life Science unit for the middle school years, so students keep meeting it between Grade 5 and Grade 11.

(Data: Life Science 47% in K-5, 7% in 6-8, 50% in 9-12) **see Key fact 6**

Coverage map ↗

WHERE THE NUMBERS COME FROM AND HOW TO READ THEM

About the data

This paper walks through what our teachers reported about how our standards are taught, from Kindergarten to Grade 12. It has eight parts: the whole school, the three grade bands, and the four subjects. Each part has three sections: Key facts, Relevant patterns, and Where do we go next? (questions for teams to take forward, and options for our part in the story).

> **These findings come from the ratings you entered as teachers.** They show what teachers reported as taught across each grade, not what students know or can do. We are sharing them back so we can see the whole picture together and decide how we move forward. Each part ends with options for our part in the story. Every item links to the data on the Vertical Alignment site.

ABOUT THE DATA

How to read this paper

> **What you are seeing.** How the ratings were made, what the three ratings mean, and the words we use.
>
> **What it adds to our story.** It helps us read every number fairly: these are teachers' reports of teaching, not measures of what students know.

This paper walks through what our teachers reported about how our standards are taught, from Kindergarten to Grade 12. It has eight parts: the whole school, the three grade bands, and the four subjects. Each part has three sections: Key facts, Relevant patterns, and Where do we go next? (questions for teams to take forward, and options for our part in the story).

Where the numbers come from. Every figure is computed from the ratings sheet of 9 November 2026 (2,166 standards, 2,709 ratings, last rating 8 November) and checked against the Vertical Alignment reports and findings decks.

The three ratings.

Rating	What it means
Taught in depth	ALL classes in the grade consistently teach, reteach, review and assess the standard
Introduced / exposed	Anything in between: some classes, or taught briefly (for example, 3 of 5 classes)

Rating	What it means
Not taught	No class in the grade teaches it

Four things to hold in mind while reading.

- The ratings are what teachers reported as taught across a whole grade. They are not a measure of what students know or can do, and they do not judge any teacher.
- "In depth" is all or nothing. Teams may have read that more or less strictly, so a low figure can reflect a stricter reading, not less teaching. Where this matters most, the paper says so. Where a team rated nothing in depth, we read it as careful use of an all-or-nothing scale, not as an absence of teaching.
- Each team rated its own subject's standards. Much literacy (vocabulary, academic language, speaking, writing, reading non-fiction) is also taught through Science and Social Studies, and that teaching may not show in the ELA ratings.
- Shares are of the main grade-level ratings (one standard in one grade). Essential Elements (the alternate standards) are counted apart.

We cannot teach everything equally. No school can take every standard to depth, and that is not the goal. The goal is to choose on purpose: which standards we teach in depth, which we introduce, and which we leave out, and to plan that proportion together across the grades. The ratings show the mix we have now, so we can decide the mix we want.

Instructional capacity. How many standards a grade actually takes to depth, introduces or leaves out in a year, worked out from the ratings (the average for one grade). It is not a target and not a judgement: it describes what our teaching time holds now, so we can decide together how to use it.

Themes. In every part, the facts, patterns and questions are grouped under the same themes, in the same order: The big picture, Strands and topics, Across grades and schools, Priority and power standards, Essential Elements, Gaps and placement, How we read the scale, Capacity and choices, Literacy across subjects and Background. Each item shows its theme, so a team can follow one theme across the whole school, its band and its subject.

How each part is laid out. Three sections, each opening with a small map that shows where you are: Key facts, then Relevant patterns, then Where do we go next? (questions to ponder, then options for our part in the story (improving the curriculum, working across grade levels, knowing the bands above and below, and a concrete action). Every pattern carries its figures in the text, and every question ends with the data behind it, in brackets and italics. The questions are invitations, not conclusions. Our strengths are named alongside the places where we might build.

ABOUT THE DATA

Method, sources and limits

> **What you are seeing.** How the ratings were made, what the three ratings mean, and the words we use.
>
> **What it adds to our story.** It helps us read every number fairly: these are teachers' reports of teaching, not measures of what students know.

Every figure in this paper was computed from the ratings sheet and cross-checked against the site. Where the site's band and K-12 views differ (progressions), the paper says which it uses. Percentages are rounded half up, as on the site.

What was read

Source	How it was used	How fully it was read
Ratings sheet, "All Standards and Ratings K-12" (9 Nov 2026)	Every figure	All 2,166 rows
Organized ratings workbook	Cross-check of totals	All tabs
20 Findings decks (band and subject)	Cross-check, questions and next-step themes	All slides and speaker notes
K-12 Reports PDF (118 pages)	Cross-check	First 80 pages scanned by search; a few read directly
7 other Reports PDFs	Cross-check of headline figures	Scanned by search, not read page by page

How terms are used

- **Rated standards.** One rating is one standard in one grade. A standard shared by several grades is counted once for each grade that rated it.
- **At least introduced.** Rated in depth or introduced.
- **No grade teaches it (confirmed).** Every grade that could teach the standard rated it not taught.
- **Not taught where placed.** The grade it was placed in rated it not taught, and other grades that could teach it were not asked.
- **Introduced, never in depth (met but not owned).** Introduced in at least one grade and in depth in none.
- **Progression finding.** An automatic check on a chain of linked standards: a step not taught, or a standard in depth while the one it builds on is not taught.
- **Taught at every step.** Every standard in the chain, across two or more grades, is at least introduced.

- **Progression.** A chain of linked standards across grades. Official links are the Coherence Map (Math), anchor standards (ELA) and the standards' own structure (Science, Social Studies). Other links are suggestions.
- **Instructional capacity.** The average number of standards one grade rated in depth, introduced and not taught in a year (all ratings divided by the number of grades), rounded.
- **Hand-over gap.** A standard taught in depth while a standard it builds on, placed in an earlier school, is rated not taught.
- **Priority and power standards.** Standards the school's own documents name: power standards in K-5, priority standards in 6-12.

Limits to keep in view

- Ratings are teachers' judgements of teaching across a grade. They are not observations and not student results.
- Grades may have read "in depth" differently. Uniform ratings across a whole grade are a reason to check how the scale was read; the paper points these out so teams can discuss them.
- The ratings are what teachers reported as taught. They do not measure what students know or can do.
- Each team rated its own subject. Literacy taught in Science and Social Studies, and Math taught in Science, may not show in the ELA and Math ratings.
- Middle school Science and high school standards were placed in grades and courses by suggestion. "Not taught" there can mean "taught in another year". 24 K-5 Social Studies standards are also not taught in the grade where they were placed.
- Band counts of progressions look only at steps inside the band, so they do not add up to the K-12 figure.
- Literacy audit figures come from the 2026 PK-5 literacy audit of weekly plans.
- Some strands have few standards in a grade, so a single rating can move a percentage a long way. Counts are given where this matters.
- Suggested links between standards (most ELA, Science and Social Studies hand-overs) have not been checked by teachers.

Status. This is a draft for the Vertical Alignment Committee. Each finding should be checked with the teams concerned before it is treated as settled. The planned review of weekly and unit plans (Part 2) will test many of these patterns against what is actually planned.